MANIPAL UNIVERSITY
JAIPUR
(University under Section 2(f) of the UGC Act)

B.TECH FIRST YEAR

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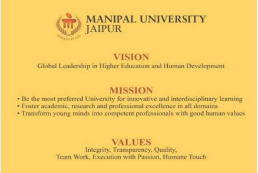

Syllabus

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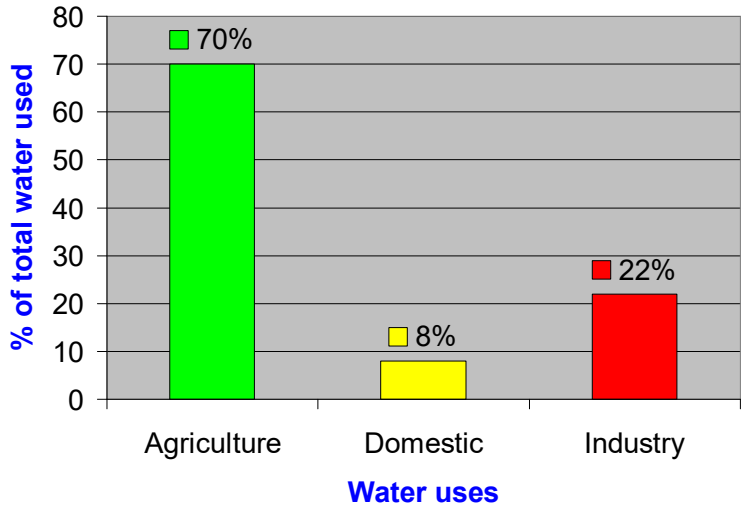
Water treatment technology. Introduction, Characteristics imparted by impurities in water, Hardness of water Degree of hardness. Determination of hardness by EDTA method. Numerical. problems. Softening of hard water: External treatment by lime soda process. Softening of hard water: Ion exchange method; Zeolite methods Softening of hard water, internal treatment by phosphate, calgon condition and colloid conditioning. Numerical problems based on lime soda process.

Water Is one of the abundant commodities in nature, but is also the most misused one

Earth is a blue planet, 80% of our planet is covered by water.

But, 97.5% of it is locked in the oceans, which is too saline to drink and other uses.

2.4% water is trapped in polar ice caps and giant glaciers, from which only 1% water is used by human for various development, industrial, agricultural and domestic purposes.



Sources of water

Surface Water:

(i) Flowing water e.g. rivers and streams

In general river water contains dissolved minerals from soil such as chlorides, sulphates, bicarbonates of sodium, calcium or magnesium, iron and organic matters derived from decomposition of plants, small particles of sand and rocks in suspensions.

(ii) Still waters e.g. lakes, ponds and reservoirs (Lowland surface drainage) Lake water has more constant chemical composition.

Underground Water: water from shallow and deep spring and wells

Rain water: It is probably the purest form of natural water

Sea Water: It is probably the most impure form of natural water

Types of Impurities Present in water

Dissolved impurities:

(a) Inorganic salts e.g.

(i) Cations: Ca^{2+}, Mg^{2+}, Na^{+}, K^{+}, Fe^{+2}, Al^{+3}, Zn^{2+}, Cu^{2+}

(ii) Anions: Cl^{-}, SO_4^{2-}, NO_3^{-}, HCO_3^{-}, F^{-}, NO_2^{-}

(b) Gases: CO_2, O_2, N_2, NH_3, H_2S

(c) Organic salts → Carboxylic acids

Suspended impurities:

(a) Inorganic e.g. clay and sand

(b) Organics e.g. oil globules, vegetable and animal matters

Colloidal impurities:

Clay and finely divided silica colloidal partials of 10^{-4} – 10^{-6} mm size

Micro-organisms:

Bacteria, Fungi, algae and other forms of animal and vegetable life

Effects of Impurities in natural water

- **Colour**
- **Taste and odour**
- **Turbidity and sediments**
- **Micro-organisms**
- **Dissolved minerals matters**
  (a) hardness
  (b) Alkalinity
  (c) Total solids
  (d) corrosion

- **Dissolved gas**
- **Silica contents**

Hardness of water

Hardness of water is originally defined as the soap consuming capacity of a water sample. The soap consuming capacity of water is mainly due to the certain salt of calcium, magnesium and other heavy metals dissolved in it.

Fe, Al., Zn (trace amount)

The soap generally consists of sodium salts of fatty acids such as Oleic acid, Palmitic acid and stearic acid. Calcium and magnesium react with the sodium salts of long chain fatty acid present in the soap to form insoluble scums of calcium and magnesium soaps.

Hardness of water

Permanent

Temporary → by boiling

$$2\,C_{17}H_{35}COONa + CaCl_2 \longrightarrow (C_{17}H_{35}COO)_2Ca + NaCl$$

(Sodium stearate) (Insoluble)

$$2\,C_{17}H_{35}COONa + MgSO_4 \longrightarrow (C_{17}H_{35}COO)_2Mg + Na_2SO_4$$

(Sodium stearate) (Insoluble)

Other metal ions such as Fe^{2+}, Mn^{2+}, Al^{3+} also contributing to hardness, but they are present in water only in traces

$CH_3(CH_2)_7CH{=}CH(CH_2)_7COOH$ **Oleic acid**

$CH_3(CH_2)_{14}CO_2H$. **Palmitic acid**

$CH_3(CH_2)_{16}CO_2H$ **stearic acid**

Temporary hardness (carbonate hardness)

Temporary hardness is caused by the presences of dissolved bicarbonate of calcium and magnesium and other heavy metal ions

Fe / Mn / Al / Zn

Temporary hardness is mostly destroyed by boiling of water. During boiling bicarbonate are decomposed in the insoluble carbonate and hydroxide, which are deposited at the bottom of the vessel.

ANIONS.

$$Ca(HCO_3)_2 \xrightarrow{\text{Heat}} CaCO_3 + CO_2 + H_2O$$

(Insoluble)

$$Mg(HCO_3)_2 \xrightarrow{\text{Heat}} MgCO_3 + 2\,CO_2 + H_2O$$

(Insoluble)

Permanent hardness (non-carbonate hardness)

This is due the presences of chlorides and sulphates of calcium, magnesium, iron and other heavy metal ions.

Hardness is expressed in terms of ***equivalent of calcium carbonate*** because it is the most **insoluble salt** that can be precipitated in water treatment.

amount of dissolved

??

$$\text{Equivalent of } CaCO_3 = \frac{\text{Mass of hardness producing substances} \times 50}{\text{Chemical equivalent of hardness producing substances}}$$

or

??

$$\text{Equivalent of } CaCO_3 = \frac{\text{Mass of hardness producing substances} \times 100}{\text{Molar mass of hardness producing substances}}$$

mol. wt = 100

Ch. eq. = 50

mol. wt.

Dissolved salt	Molar Mass	Chemical equivalent	Multiplication factor for converting into equivalent of $CaCO_3$
$Ca(HCO_3)_2$	162	81	100/162
$Mg(HCO_3)_2$	146	73	100/146
$CaSO_4$	136	68	100/136
$CaCl_2$	111	55.5	100/111
$MgSO_4$	120	60	100/120
$MgCl_2$	95	47.5	100/95
$MgCO_3$	84	42	100/84

Units of Hardness:

ppm
mg/l
Degree Clarke (°Cl)
Degree French (°Fr)

A Clark degree (°Clark) or English degrees (°e or e) is defined as one grain (64.8 mg) of $CaCO_3$ per Imperial gallon (4.55 litres) of water, equivalent to 14.254 ppm.
(mg/L)

A French degree (°fH or °f) is defined as 10 mg/L $CaCO_3$, equivalent to 10 ppm.

Estimation of hardness: Hardness are usually determined by two methods

Soap solution methods:-

soaps gives lather with hard water only after sufficient quantity of soaps is added to precipitate all the hardness causing metal ions present in water

$$\underset{\text{(Sodium stearate)}}{2\,C_{17}H_{35}COONa} + CaCl_2 \text{ or } MgCl_2 \longrightarrow \underset{\text{(Insoluble)}}{(C_{17}H_{35}COO)_2Ca \text{ or } Mg} + NaCl$$

$$\underset{\text{(Sodium stearate)}}{2\,C_{17}H_{35}COONa} + MgSO_4 \text{ or } CaSO_4 \longrightarrow \underset{\text{(Insoluble)}}{(C_{17}H_{35}COO)_2Mg \text{ or } Ca} + Na_2SO_4$$

After precipitation of all the hardness causing ions present in water, further addition of soap gives lather

$CaCO_3 + HCl \rightarrow CaCl_2 + H_2O + CO_2$

The solutions required are:

1.A standard solution of $CaCl_2$ prepared by dissolving 0.2 gm of pure $CaCO_3$ in a small quantity of dilute hydrochloric acid. The solution is then evaporated, to dryness to remove excess acid. The residue is thereafter dis-solved in one liter of distilled water.

Na. stearate 1 l of 80% EtOH

2. **Standard soap solution**: 100 gms of pure dry soap is dissolved in one liter of 80 per cent alcohol. This solution is diluted with such quantity of 80 per cent alcohol that 6.4 ml when added to 20 ml of standard $CaCl_2$ solution produces a pertinent lather.

Procedure:

- 50 ml of water under test is taken in a stoppered bottle and to it is added standard soap solution by burette 0.2 ml at a time.
- The mixture is shaken vigorously after each addition until lather formation starts.
- Then the addition of soap. solution is made at the rate of 0.1 ml at a time till a lather is obtained which persists for 5 minutes when the bottle is kept aside.
- The total soap solution added is read from the burette. The hardness as $CaCO_3$ is then found out from a standard table.

Procedure:

For the determination of permanent hardness the water is boiled and then treated with a measured excess of standard Na_2CO_3 solution, and the filtrate treated with a standard acid using Methyl Orange as indicator, and the volume of Na_2CO_3. originally added, minus that left over, is the volume actually consumed in the precipitation of alkaline earth salts, and its $CaCO_3$ equivalent gives the permanent hardness:

$CaSO_4 + Na_2CO_3 = CaCO_3 + Na_2SO_4$
$MgCl_2 + Na_2CO_3 = MgCO_3 + 2NaCl$
i.e., 1 ml of 0.1N. Na_2CO_3 = 0.005 gm $CaCO_3$

Total hardness
− Permanent hardness
Temporary hardness

EDTA Method:-

Mg^{2+} / Ca^{2+}

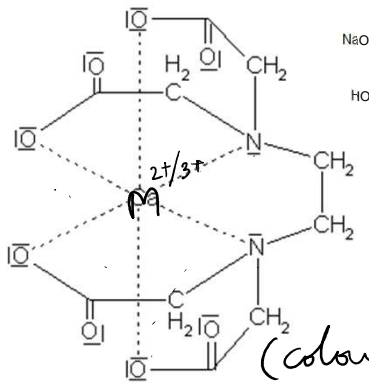
Hard water

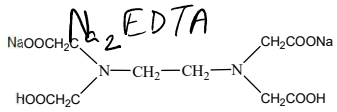
+

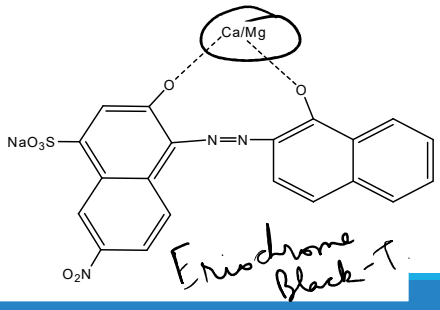
(blue) **EBT Indicator**

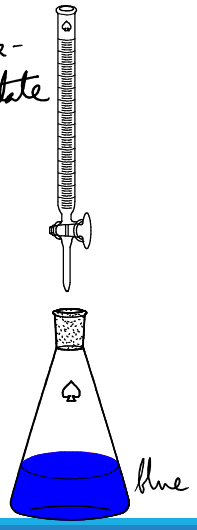
+

10 pH buffer solution









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WATER SOFTENING

Lime : $Ca(OH)_2$

Soda : Na_2CO_3

removal of hardness

- Hardness is?...

Primarily due to Ca, Mg, as well as Fe, Mn, Sn, Al

How is Softening done?...

Precipitation of Ca and Mg, or

Ion exchange of Ca^{2+} / Mg^{2+} with ion such as Na

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Hardness in 300-500 mg/L as $CaCO_3$ range considered excessive

Even > 150 mg/L may result in consumer objection

60-120 mg/L as $CaCO_3$ is considered a moderate amount

Hardness < 30 ppm

Disadvantages of H.W.

high soap consumption

scaling in heating vessels and pipes

Huge q.

Lime-Soda process

In this process, all the soluble hardness-causing impurities are converted into insoluble precipitates which may removed by setting and filtration .

In the lime soda process, calculated amount of lime $Ca(OH)_2$ and soda Na_2CO_3 is added in the hard water. The soluble calcium and magnesium salts in water are converted into insoluble compound such as calcium carbonate and magnesium hydroxide which may removed by setting and filtration.

(i) Lime removes the temporary hardness:

$$Ca(HCO_3)_2 + Ca(OH)_2 \longrightarrow 2\,CaCO_3\downarrow + 2\,H_2O$$

$$Mg(HCO_3)_2 + 2\,Ca(OH)_2 \longrightarrow Mg(OH)_2\downarrow + CaCO_3\downarrow + H_2O$$

Ca^{2+}/Mg^{2+} Temp.

(ii) Lime removes the permanent magnesium hardness:

$$MgCl_2 + Ca(OH)_2 \longrightarrow Mg(OH)_2\downarrow + CaCl_2$$

$$MgSO_4 + Ca(OH)_2 \longrightarrow Mg(OH)_2\downarrow + CaSO_4$$

Mg^{2+} Perm

(iii) Lime removes the dissolved iron and aluminum salts:

$$FeSO_4 + Ca(OH)_2 \longrightarrow Fe(OH)_2 \downarrow + CaSO_4$$

$$Al_2(SO_4)_3 + 3\,Ca(OH)_2 \longrightarrow 2\,Al(OH)_3 \downarrow + 3\,CaSO_4$$

Fe^{2+}/Al^{3+} Perm.

(iv) Lime removes free mineral acids:

$$2\,HCl + Ca(OH)_2 \longrightarrow CaCl_2 + H_2O$$

$$H_2SO_4 + 3\,Ca(OH)_2 \longrightarrow CaSO_4 + H_2O$$

H^+ Ca^{2+} Perm.

(v) Lime removes dissolved CO2 and H2S:

$$CO_2 + Ca(OH)_2 \longrightarrow CaCO_3 \downarrow + H_2O$$

$$H_2S + 3\,Ca(OH)_2 \longrightarrow CaS \downarrow + 2\,H_2O$$

Dissolved CO_2 gas H_2S.

(vi) Soda removes all calcium permanent hardness:

$$CaCl_2 + Na_2CO_3 \longrightarrow CaCO_3 \downarrow + NaCl$$

$$CaSO_4 + Na_2CO_3 \longrightarrow CaCO_3 \downarrow + Na_2SO_4$$

Now the 100 parts by mass of CaCO3 are equivalent to : (i) 74 part of $Ca(OH)_2$ and (ii) 106 parts of Na_2CO_3

$Ca(OH)_2$

Lime required for softening:

$$= \frac{74}{100}[\text{Temp. } Ca^{2+} + 2\times\text{Temp. } Mg^{2+} + \text{Perm. } (Mg^{2+} + Fe^{2+} + Al^{3+}) + CO_2 + H^+ (HCl \text{ or } H_2SO_4) + HCO_3^-]$$

Na_2CO_3

Soda required for softening:

$$= \frac{106}{100}[\text{Perm. } (Ca^{2+} + Mg^{2+} + Fe^{2+} + Al^{3+}) + CO_2 + H^+ (HCl \text{ or } H_2SO_4) - HCO_3^-]$$

Cold lime soda process:

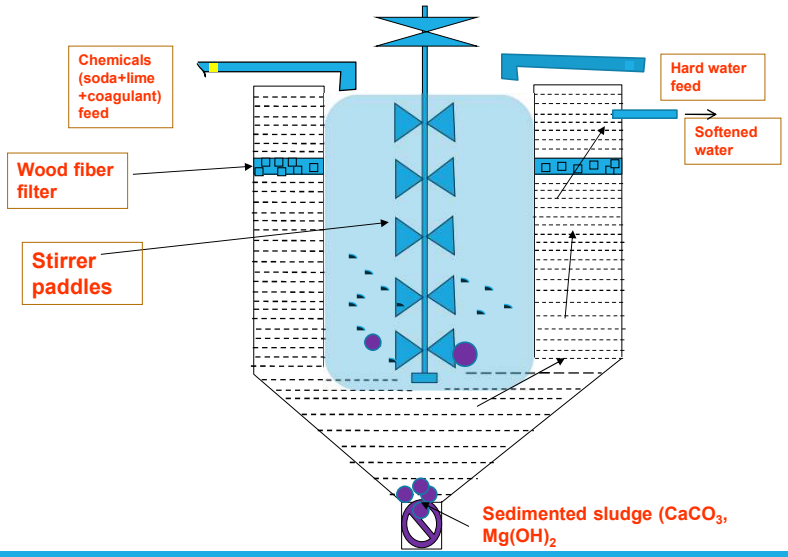
Calculated quantity of lime and soda are mixed with water at room temperature the precipitates formed are finely divided, so they do not settle down easily. Therefore, It is essential to add small amounts of coagulants (like alum, aluminum sulphate, sodium aluminates *etc*).

Use of sodium aluminate as coagulant also helps the removal of silica as well as oil if present in water..

$$NaAlO_2 + 2H_2O \longrightarrow NaOH + Al(OH)_3$$

$$Al_2(SO_4)_3 + 3Ca(HCO_3)_2 \longrightarrow 2Al(OH)_3 + 3CaSO_4 + 6CO_2$$

Continuous cold lime soda softener



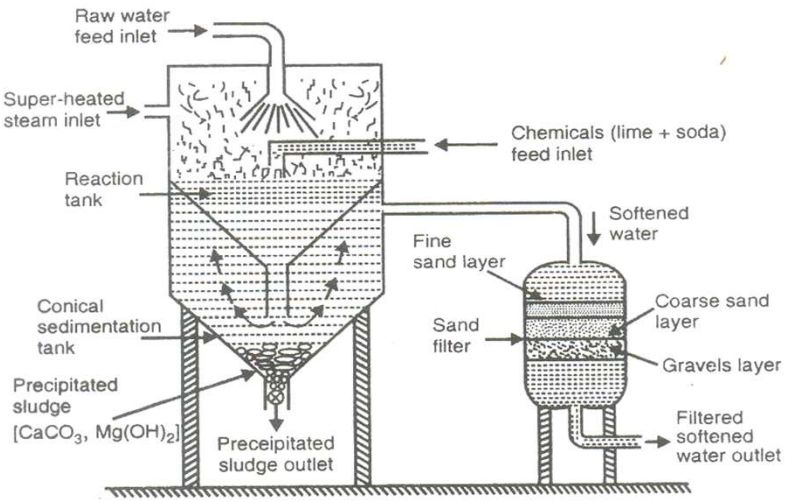
Hot lime soda process:

Calculated quantity of lime and soda are mixed with water at 80^{0}C.

Advantages of high temperature:

- The reaction proceeds faster.
- Softening capacity is increased.
- No coagulant needed as the precipitate and sludge formed settle down rapidly.
- Much of the dissolved gases are driven out of water.
- Viscosity of soften water is lower, so filtration of water becomes easier.
- This process produces water of comparatively low residual hardness 15 to 30 ppm.

Continuous Hot Lime Soda Process



15 - 30 ppm

$Ca(OH)_2$ / Na_2CO_3

Advantages of lime soda process:

- It is very economical.
- If the process is combined with sedimentation/coagulation lesser amounts of coagulants shall be needed.
- The process increases the pH of the treated .
- In addition to the removal of hardness, the quantities of minerals in the water are reduced.
- Due to alkaline nature of treated water amount of pathogenic bacteria in water is considerably reduced.

Disadvantages of lime soda process:

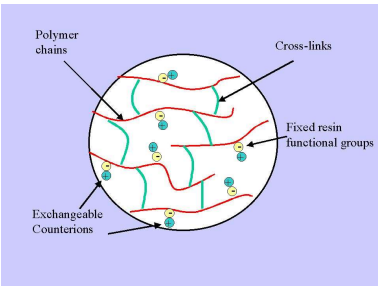
stoichiometric amount

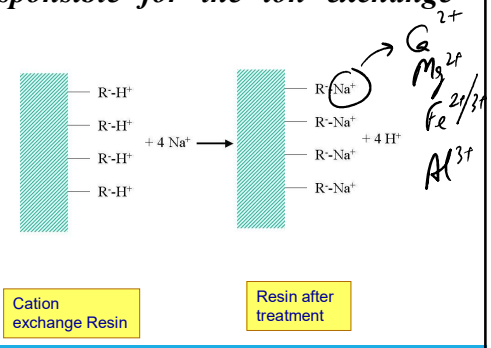
- For efficient and economical softening, careful operation and skilled supervision is required.
- Disposal of large amount of sludge poses a problem.
- This can remove hardness up to 15 ppm which is not good for boilers.

WATER SOFTENING

Ion exchange or deionization or demineralization process: Ion-exchange resins are widely used in different separation, purification, and decontamination processes. The most common examples are water softening and water purification.

Ion-exchange resins are insoluble, cross linked, long chain organic polymer with micro porous structure and the functional groups attached to the chains are responsible for the ion exchange properties.



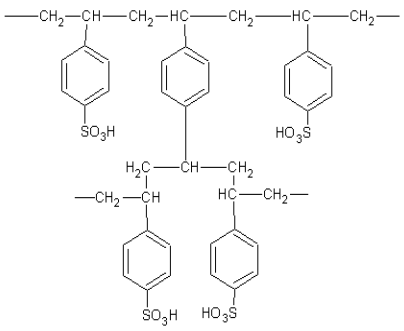


Ca^{2+} Mg^{2+} $Fe^{2+/3+}$ Al^{3+}

Resins containing acidic functional groups (-COOH, $-SO_3H$) are capable of exchanging their H^+ ions with other cations which comes in their contacts, known as a **cation exchange resins** (**RH^+**).

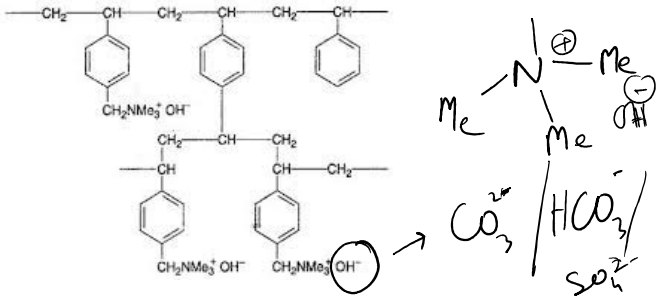
e.g. *styrene-divinyl benzene copolymer*, which on sulphonation and carboxylation, become capable to exchange their hydrogen ions with the **cations (Metal ions)** in the water.

Na^+



Resins containing basic functional groups ($-NR_3^+OH^-$) are capable of exchanging their anions with other anions which comes in their contacts, known as an **Anion exchange resins** (**ROH^-**).

They are styrene divinyl benzene or amine-formaldehyde copolymerization which contains quaternary ammonium or quaternary phosphonium or tertiary sulphonium groups as an integral part of the resin matrix. These after treated with dil NaOH becomes capable of exchanging their OH^- ions with anions of water.



Process:- The hard water is passed first through cation exchange column, which remove all the cations from it and **equivalent amount of hydrogen** ions are released from this column to water:

$-COOH$ $\qquad$ $-COO$ $-COO$ Ca

$$2RH^+ + Ca^{2+} \longrightarrow R_2Ca^{2+} + 2H^+$$

$$2RH^+ + Mg^{2+} \longrightarrow R_2Mg^{2+} + 2H^+$$

After cation exchange column the hard water is passed through anion exchange resin column, which remove all the anions from it and **equivalent amount of hydroxide** ions are released from this column to water :

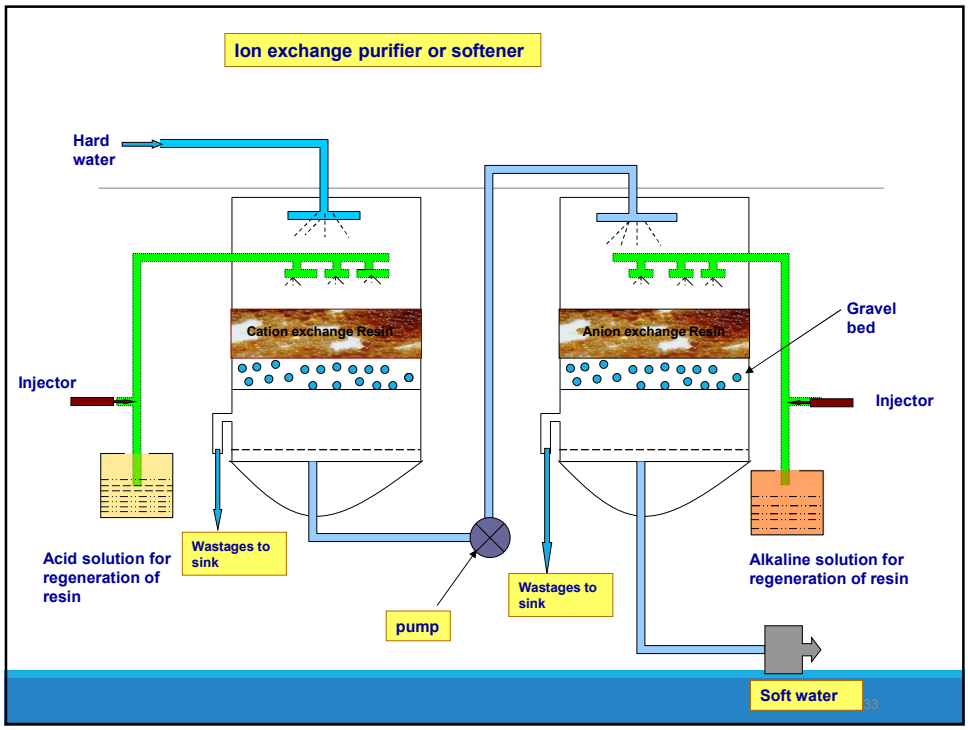
$$ROH^- + Cl^- \longrightarrow RCl^- + OH^-$$

$$2ROH^- + SO_4^{2-} \longrightarrow R_2SO_4^{2-} + 2OH^-$$

$$H^+ + OH^- \longrightarrow H_2O$$

Thus water coming out from the exchange is free from cations as well as anions.

Ion free water is known as deionized or demineralised water.


Ion exchange purifier or softener
Hard water
Cation exchange Resin
Anion exchange Resin
Gravel bed
Injector
Injector
Wastages to sink
Wastages to sink
Acid solution for regeneration of resin
Alkaline solution for regeneration of resin
pump
Soft water
33

33

34

Regeneration:

Cation exchange column is regenerated by passing a solution of dil HCl or dil H_2SO_4. The regeneration can be represented as

R_2M^{2+}

$$R_2Ca^{2+} + 2H^+ \longrightarrow 2RH + Ca^{2+}$$

Exhausted anion exchange column is regenerated by passing a solution of dil. NaOH. The regeneration can be represented as

$$R_2SO_4^{2-} + 2OH^- \longrightarrow 2ROH + SO_4^{2-}$$

Advantages:

$M^{2+/3+}$ / $A^{-/2-}$

- Can be used to soften highly acidic or alkaline waters.; pH of H_2O.
- It produces water of very low hardness.

Disadvantages:

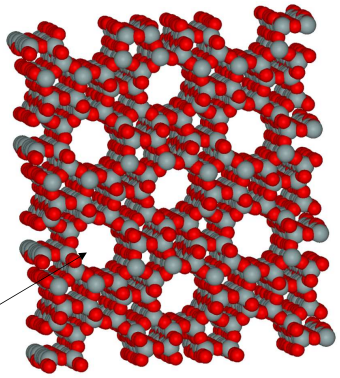
Colloids (silica) ↓ suspended materials

- The equipment is costly
- Expensive chemicals are needed
- Output of the process is reduced if water contains turbidity.(turbidity must be below 10 ppm)

Zeolite (Permutit) method of Softening of water

Zeolite is a Hydrated Sodium Alumino Silicate (HSAS), capable of exchanging reversibly its sodium ions for hardness producing ions in water.

The general chemical structure of zeolite is given below $Na_2O.Al_2O_3.xSiO_2.yH_2O$ (x = 2-10 and y = 2-6)

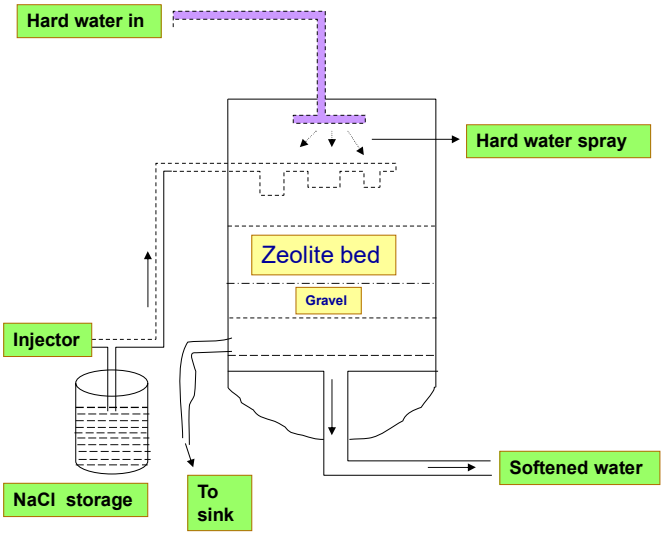


Micro pores of Zeolite

Porous Structure of zeolite

- **Porosity or cavity size of synthetic zeolite structures can be controlled by varying the Si/Al ratio**
- **Ion-exchange process of zeolite structure is associated with sodium ions**

Zeolite softener



Process of softening by Zeolite method

For the purification of water by the zeolite softener, hard water is passed through the zeolite bed at a specified rate. The hardness causing ions such as Ca^{2+}, Mg^{2+} are retained by the zeolite bed as CaZe and MgZe respectively; while the outgoing water contains sodium salts. The following reactions takes place during softening process

To remove temporary hardness

$$Na_2Ze + Ca(HCO_3)_2 \longrightarrow CaZe + 2NaHCO_3$$

Hardness

$$Na_2Ze + Mg(HCO_3)_2 \longrightarrow MgZe + 2NaHCO_3$$

To remove permanent hardness

$$Na_2Ze + CaCl_2 \longrightarrow CaZe + 2NaCl$$

$$Na_2Ze + MgSO_4 \longrightarrow MgZe + Na_2SO_4$$

$$R_2Ca^{2+} \xrightarrow{dil\ HCl/H_2SO_4} R_2H^+$$

$$R_2SO_4^{2-} \xrightarrow{dil\ NaOH} R\ OH^-$$

$$Ca\ Ze \xrightarrow{NaCl} Na_2Ze$$

Regeneration of Zeolite Bed

$$CaZe\ (or)\ MgZe + 2NaCl \longrightarrow Na_2Ze + CaCl_2\ (MgSO_4)$$

Used Zeolite | 10% brine solution | Regenerated Zeolite | Washings drained

Limitations of Zeolite process

1. If the water is turbid: then the turbidity causing particles clogs the pores of the Zeolite and making it inactive
2. The ions such as Mn^{2+} and Fe^{2+} forms stable complex Zeolite which can not be regenerated that easily as both metal ions bind strongly and irreversibly to the zeolite structure.
3. Any acid present in water (acidic water) should be neutralized with soda before admitting the water to the plant.

Advantages of Zeolite process

1. Soft water of 10-15 ppm can be produced by this method
2. The equipment occupies less space
3. No impurities are precipitated, hence no danger of sludge formation in the treated water
4. It does not require more time and more skill

Disadvantages of Zeolite process

1. Soft water contains more sodium salts than in lime soda process
2. It replaces only Ca^{2+} and Mg^{2+} with Na^{+} but leaves all the other ions like HCO_3^- and CO_3^{2-} in the softened water (then it may form $NaHCO_3$ and Na_2CO_3 which releases CO_2 when the water is boiled and causes corrosion)
3. It also causes caustic embrittlement when sodium carbonate hydrolyses to give NaOH

Internal treatment:

< 15 ppm of hardness

In this process, an ion is prohibited to exhibit its original character by complexing or converted into other more soluble salt by adding appropriate reagent.

An internal treatment is accomplished by adding a proper chemical to water either:

(a) To precipitate the scale forming impurities in the form of sludge, which can be removed.

(b) To convert them into compounds, which will stay in dissolved form in water and thus do not cause any harm.

Important Internal Treatments are:

$Na_3PO_4 \rightleftharpoons 3Na^+ + PO_4^{3-}$

$NaH_2PO_4 \rightleftharpoons Na^+ + H_2PO_4^-$

$Na_2HPO_4 \rightleftharpoons 2Na^+ + HPO_4^{2-}$

Phosphate conditioning:

Scale formation can be avoided by adding sodium phosphate which reacts with hardness of water forming non-adherent and easily removable soft sludge of calcium and magnesium phosphate.

$$3CaCl_2 + 2Na_3PO_4 \longrightarrow Ca_3(PO_4)_2 + 6NaCl$$

The choice of salt depends upon the alkalinity of the boiled water because calcium cannot be precipitated below a pH 9.5.

- Trisodium (Na_3PO_4) phosphate is most suitable for treatment when alkalinity is low
- Disodium (Na_2HPO_4) phosphate is used when the water alkalinity is sufficient.
- Monosodium (NaH_2PO_4) phosphate is used when the alkalinity of boiler water is too high.

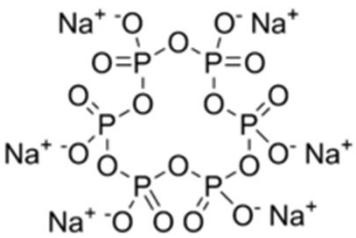
Calgon conditioning:

It involves adding calgon (Sodium hexa-meta phosphate $(NaPO_3)_6$ to boiler water. It prevents the scale and sludge formation by forming soluble complex compound with $CaSO_4$.

$$[Na_6(PO_3)_6] \longrightarrow 2Na^+ + [Na_4P_6O_{18}]^{2-}$$

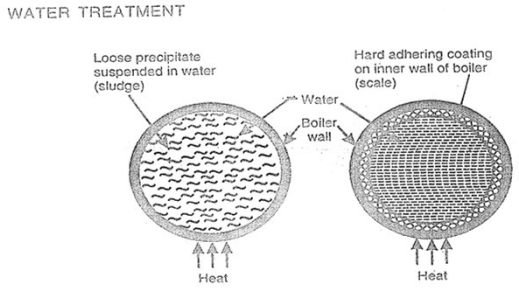
$$2CaSO_4 + [Na_4P_6O_{18}]^{2-} \longrightarrow [Ca_2P_6O_{18}]^{2-} + 2Na_2SO_4$$

soluble complex ion



Scales and Sludges Formation In Boilers:

In boilers, water evaporates continuously and the concentrations of the dissolved salts increases progressively. when concentrations of dissolved salts reach saturation point, they form precipitates in form of loose and slimy, it is called **sludge**. On the other hand, if the precipitate matter form a hard adhering crust/coating on the inner walls of the boiler, it is called **scale.**



Scales are hard deposit, which stick very firmly to the inner surface of boiler. Scales are very difficult to remove.

Formation of scales may be due to

(1) Decomposition of calcium bicarbonate: (In low-pressure boilers)

$$Ca(HCO_3)_2 \rightarrow CaCO_3 + H_2O + CO_2$$

Scale composed mainly from calcium carbonate is soft and is the main cause of scale formation in low pressure boiler. But in the high pressure boiler, calcium carbonate is soluble.

(In high-pressure boilers)

$$CaCO_3 + H_2O \rightarrow Ca(OH)_2 \text{ (soluble)} + CO_2$$

(2) Deposition of Calcium Sulphate:

Solubility of calcium sulphate in water decrease with rise of temperature. Hence $CaSO_4$ gets precipitated as hard scale on the heated portions of the boiler.

(3) Hydrolysis of magnesium salts:

Dissolved Mg salts undergo hydrolysis forming magnesium hydroxide precipitate, which forms a soft type of scale

$$MgCl_2 + 2H_2O \rightarrow Mg(OH)_2 + 2HCl$$

(Scale)

Permanent hardness

(4) Presence of silica:

SiO_2 even present in small quantities, deposits as calcium silicate ($CaSiO_3$) and /or magnesium silicate ($MgSiO_3$). These deposits stick on the inner side of the boiler surface and are very difficult to remove.

Disadvantages of scale formation

- **Wastage of fuel:** Scales have a low thermal conductivity, so that rate of heat transfer from boiler to inside water is greatly decreased
- **Lowering of boiler safety:** Due to scale formation, over heating of boiler is done, which causes distortion of boiler tube.
- **Decrease in efficiency:** Scales may deposit in the valve and condensers of the boiler and choke them partially.
- **Danger of explosion:** When thick scales crack, the water comes suddenly in contact with over heated iron plates.

Removal of scales

- With the help of scraper or piece of wood or wire brush.
- By giving thermal shocks, if they are brittle.
- By dissolving them by adding chemicals, (5-10% HCl, EDTA) if they are adherent and hard.
- By frequent blow-down operation, if the scales are loosely adhering.

Prevention of scales formation

(1) External Treatment: Includes efficient 'softening of water'
(i.e., removing hardness-producing constituents of water)

(2) Internal Treatment: Accomplished by adding a proper chemical to the boiler water either:

Important internal conditioning/treatment methods:

(i) Colloidal conditioning: Scale formation can be avoided by adding organic substances like kerosene, tannin, agar-agar (a gel) etc., which get coated on over the scale forming precipitate, thereby, yielding non-sticky and loose deposits.

(ii) Phosphate Conditioning: Na_3PO_4 / Calgon.

(iii) Carbonate Conditioning: In low pressure boilers, scale-formation can be avoided by adding sodium carbonate to boiler water.

$$CaSO_4 + Na_2CO_3 \rightarrow CaCO_3 + Na_2SO_4$$

(iv) Calgon (**cal**cium **gon**e) **Conditioning:**. sodium hexametaphosphate

(v) Treatment with sodium aluminate ($NaAlO_2$): sodium aluminate gets hydrolyzed yielding NaOH and a gelatinous of aluminum hydroxide

$$NaAlO_2 + 2H_2O \rightarrow NaOH + Al(OH)_3$$

Sod.meta-aluminate (Gelatinous ppt.)

Theses sodium hydroxide precipitates some magenisium

$$MgCl_2 + 2NaOH \rightarrow Mg(OH)_2 + 2NaCl$$

The flocculent precipitate of $Mg(OH)_2$ and $Al(OH)_3$ produced inside the boiler, entraps finely suspended and colloidal impurities, including oil drops and silica.

(vi) Electrical conditioning: Sealed glass bulbs, containing Hg connected to a battery, are set rotating in the boiler. when water boils, mercury bulbs emit electrical discharges which prevents scale forming particles to adhere/stick together to form scale.

(vii) Radioactive conditioning: Tablets (radioactive salts) are placed inside the boiler water. Energy radiations emitted by these salts prevent scale formation